

Rishabh Prajapati

Patna, Bihar, India

+91-7860634479 | prajapatirishabh071@gmail.com

GitHub | Portfolio | LinkedIn

SUMMARY

AI/ML undergraduate at IIT Patna with hands-on experience shipping 5+ generative-AI systems - RAG pipelines, multi-agent LLM applications, and OCR-driven extraction pipelines - using Python, LangChain, FastAPI, and vector databases. As an AI Intern at Aagento, automated engineering-drawing validation for railway infrastructure, cutting manual review effort by 40% and reducing design-check turnaround time by 50%.

EDUCATION

●**B.S, Computer Science and Data Analytics** - Indian Institute of Technology, Patna

Patna, Bihar, India | 2023 - 2027

EXPERIENCE

●**AI Intern** - Aagento

On-site | May 2026 - Jul 2026

- Engineered a **GAD extraction and validation system** using **MinerU, EasyOCR, PaddleOCR, Tesseract OCR, and MiniCPM-V**, automating extraction of engineering data from railway bridge drawings and digitizing legacy GAD drawings.
- Architected **5+ FastAPI-based rule engines** covering **Hydrology, Foundation, Span Arrangement, Level Computation, and Structural Code** validation, reducing design-check turnaround time by **50%**.
- Implemented **3+ RAG systems** for semantic search, document retrieval, and question answering, improving retrieval accuracy by **30%** over keyword-based search.
- Benchmarked **5 LLMs** - MiniCPM, Kimi, Ministral, Llama-3.3 Nemotron Super, and Nemotron Ultra - for **GAD data extraction and validation** across **100+ engineering documents**, evaluating extraction accuracy and selecting the best-performing model for production deployment.

PROJECTS

●**Enterprise Hybrid RAG - Hybrid Search & Retrieval Pipeline (GitHub)**

- Built a production-grade RAG system integrating **Qdrant + BM25 hybrid retrieval, Reciprocal Rank Fusion, and Cross-Encoder reranking**, with configurable chunking and cosine-based deduplication.
- Implemented **citation entailment verification, confidence scoring, and graceful fallback mechanisms** using LangChain LCEL and FastAPI for grounded question answering.
- **Technologies:** Python, FastAPI, LangChain, Qdrant, BM25, RRF, Cross-Encoder, Docker, Angular

●**Lumi AI - Second Brain (RAG-based Knowledge Assistant) (GitHub)**

- Built a second-brain assistant powered by LangGraph agent workflows and a FastAPI backend, using PostgreSQL pgvector for semantic-search RAG, Tavily for live web search, and a multilingual (15+ language) voice recognition and synthesis pipeline, served through a glassmorphism Angular UI.
- **Technologies:** FastAPI, LangGraph, LangChain, PostgreSQL, pgvector, OpenRouter, Tavily, Sentence-Transformers, Angular, Python

●**Medical ChatBot (RAG-based Healthcare Assistant) (GitHub)**

- Developed a medical chatbot on a RAG architecture with Gemini LLMs and LangChain, indexing 1,000+ medical documents in Pinecone and serving answers through a Flask web interface.
- **Technologies:** RAG, Gemini LLMs, LangChain, LangSmith, Pinecone, Flask, Python, HTML/CSS, JavaScript

●**AI Operations Assistant (Multi-Agent LLM System) (GitHub)**

- Designed a multi-agent AI system powered by LangGraph and Gemini LLM that translates natural-language commands into executable actions across Planner, Executor, and Verifier agents, integrating GitHub and OpenWeather APIs with Pydantic-validated I/O.
- **Technologies:** Python, Streamlit, LangChain, LangGraph, Gemini API, GitHub API, OpenWeather API, Pydantic

TECHNICAL SKILLS

- **Programming:** Python, SQL
- **Machine Learning & NLP:** Supervised Learning, Unsupervised Learning, Deep Learning, CNN, RNN, LSTM, Natural Language Processing
- **Generative AI:** RAG, LangChain, LangGraph, Gemini API, Vector Databases (Pinecone, FAISS)
- **Data Analysis:** Exploratory Data Analysis, Feature Engineering, Statistical Analysis
- **Tools & Platforms:** Git, MLflow, Jupyter Notebook, Hugging Face, Docker
- **Deployment:** Streamlit, Flask, FastAPI, REST APIs

POSITIONS OF RESPONSIBILITY

●**Member**, Technology Club under the Student Technical Council, IIT Patna - Mar 2025 - Dec 2025

- Evaluated 30+ AI/ML/DL projects during the Algnition Hackathon, judging on innovation, technical depth, and real-world applicability.

CERTIFICATIONS

- Data Science** - Guvi HCL (Jun 2025) (Verify)
- Data Visualization** - Kaggle (Jan 2025) (Verify)
- PyTorch Ultimate 2024: From Basics to Cutting-Edge** - Udemy (Dec 2024) (Verify)
- Pandas** - Kaggle (Dec 2024) (Verify)